

Task 1: Project Phase-I Summary

Students must briefly summarise the Phase I project. Students are required to:

- Restate the selected problem, dataset, target variable, and prediction objective.
- Summarise the dataset source, number of records, number of features, and main data characteristics.
- Briefly describe the main data-quality issues identified in Phase I.
- Explain why the problem is suitable for an LCS-based solution.

Task 2: Original LCS System on raw Dataset

Students must run an original LCS system on the raw or original dataset before making improvements. Students are required to:

- Select an appropriate LCS variant, preferably eLCS.
- Use the original LCS code without algorithmic modification.
- Run the original LCS system on the raw or minimally processed dataset.
- Report the selected LCS parameters.
- Record the baseline performance results.
- Explain any minimal processing required to make the dataset compatible with the LCS code.

Task 3: Data Preprocessing and Feature Engineering

Students must improve the quality of the dataset before developing the improved system and make it suitable for working with LCS. Students are required to:

- Address missing values, duplicate records, invalid values, inconsistent categories, and incorrect data types.
- Investigate and handle outliers where appropriate.
- Encode categorical variables and scale, discretise, or transform numerical variables where required.
- Address class imbalance where relevant.
- Apply feature engineering, feature selection, or dimensionality reduction where justified.
- Clearly explain how preprocessing is expected to improve LCS performance.
- Ensure that preprocessing avoids data leakage.

Task 4: Improved LCS-Based System

Students must improve the LCS code, LCS configuration, or machine learning pipeline to develop a novel improved system. Possible improvements include:

- Hyperparameter tuning.
- Improved preprocessing designed for LCS performance.
- Feature selection or dimensionality reduction.
- Rule pruning or rule compaction.
- Modified LCS algorithm.
- Class-imbalance handling.
- Cost-sensitive learning.

Students are required to:

- Clearly identify what was changed compared with the original LCS system.
- Justify why the change may improve prediction accuracy or model quality.
- Provide a brief algorithmic explanation, pseudocode, or flowchart of the improved system.
- Compare the improved system with the original LCS baseline.

Task 5: Experiments

Students must use a suitable and justified validation strategy. Students are required to:

- Use either an *80/20 train-test split* or *k-fold cross-validation*.
- Justify the selected validation approach.

Students must evaluate results using appropriate metrics and statistical tests.

Students are required to:

- Select evaluation metrics suitable for the selected problem.
- For classification, consider accuracy, precision, recall, F1-score, balanced accuracy, ROC-AUC, PR-AUC, and confusion matrix.
- For regression, consider MAE, MSE, RMSE, MAPE, R^2 , and residual analysis.
- Justify the selected metrics.
- Apply at least one suitable statistical test to compare model performance.
- Interpret whether performance differences are statistically significant.

Task 6: Model Comparison

Students must compare the proposed improved LCS-based system with the original LCS and at least three conventional machine learning models. The comparison must include:

- Original LCS on the raw or minimally processed dataset.
- Original LCS on the preprocessed dataset.
- Improved LCS-based system on the preprocessed and/or feature-engineered dataset.

- At least three conventional non-deep-learning machine learning models.

Conventional models may include Random Forest, Support Vector Machine, Decision Tree, Logistic Regression, Naive Bayes, k-Nearest Neighbours, Gaussian Process, or other appropriate non-deep-learning models. Deep-learning models are not permitted for the required model comparison.

Task 7: Interpretation of Results

Students must include a separate section on result interpretation and Explainable AI. Students are required to:

- Interpret the main findings from the model comparison.
- Explain important LCS rules, feature conditions, or classifier patterns.
- Use at least *three sample LCS rules/classifiers* to discuss the decision-making process.
- Discuss how LCS rule-based outputs support interpretability.
- Discuss whether the explanations are trustworthy and practically meaningful.

Task 8: Discussion, Limitations, and Future Work

Students must critically discuss the results and the responsible use of the developed system. Students are required to discuss:

- Reasons for performance improvement or lack of improvement.
- Practical meaning of the results, not only numerical performance.
- Limitations due to data quality, dataset size, class imbalance, noise, missingness, or feature relevance.
- Future work, such as improved LCS variants, better preprocessing, external validation, or domain expert review.

Deliverables

Each group must submit:

- Final report: 5,000--6,500 words, excluding references and appendices.
Source code or Jupyter Notebook.
GitHub repository link and commit-history evidence.
Contribution statement, including a brief individual reflection from each group member.
Cleaned and feature-engineered dataset, or clear dataset reproduction

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instructions.

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